

TEZPUR UNIVERSITY
Spring Semester Mid-term Examination, 2018
CO205: Formal Languages & Automata (B.Tech)

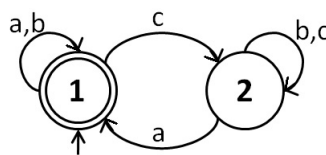
Full Marks : 40

Time : 75 mins

1. Give *one word* answers. [2]
 - (a) The process of converting an NFA to a DFA is called _____.
 - (b) A string of variables and terminals that can be derived from the start variable in a Context Free Grammar is called _____.
2. What is *ambiguity* in a Context Free Grammar? What is an inherently ambiguous grammar? Give examples. [3+3+3]
3. (a) State the Pumping lemma for regular languages. [3]
 - (b) Explain why the language L is non-regular. [4]

$$L = \{w | w \in \Sigma^*, \Sigma = \{0, 1, +, =\} \text{ and } \\ w \text{ is of the form } x = y + z \text{ where } \\ x, y, z \text{ are binary integers and } x \text{ is the sum of } y \text{ and } z \\ \}$$

4. Construct regular expressions for the following regular languages:
 - (a) $L = \{w \in \{a, b, c\}^* | w \text{ has a substring with odd number of } a\text{'s}\}$ [3]
 - (b) L is the language of the following DFA. Give textual explanation and examples of two strings accepted by both the DFA and the regular expression [6+2]



5. (a) What is ε -closure of a state? [3]
 - (b) Describe the table filling algorithm for DFA minimization. How can it be used for determining equivalence of two DFAs [6+2]